



機電整合與虛擬實境實驗 Mechatronics and XR Lab 2026

Lab 2

馬達驅動、感測器與中斷

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Today's Six Tasks

From motor drivers to interrupts — filling in the gaps



1 Basic Task 2-1 DC Motor Rotation

Rotating a DC motor with Arduino



2 Advanced Task 2-1 DC Motor Control with a Variable Resistor

Using a variable resistor to change the direction and speed of rotation of a DC motor



3 Basic Task 2-2 Servo Motor Control

Controlling the angle of the servo motor

Bonus Task: Angle Value Input through the Serial Monitor



4 Advanced Task 2-2 Servo Motor Control with an Ultrasonic Sensor

Controlling the angle of the servo motor according to the measured distance



5 Basic Task 2-3 External Interrupt

External interrupt with Arduino



6 Advanced Task 2-3 External Interrupt vs Polling

Comparing external interrupt and blocking delay with push buttons and LEDs

The First Lab Where Things Can Actually Burn

From Lab 2, you'll need to do a lot more search and investigation by yourselves to complete the tasks.



Be careful with motor circuits

If it's hot to the touch or smells burnt, cut the power immediately



GND must be shared

Even with a separate motor power supply, its GND must connect to the Arduino's



Only pins 2/3 support interrupts

Arduino Uno has just these two external-interrupt-capable pins



Task Check & Submission Rules

You must complete the task check, and upload your files for Advanced Tasks



For every task

- **All tasks** must be checked by the TAs — not just Advanced Tasks
- Call the TA over and briefly explain your sketch and circuit yourself
- The TA may ask follow-up questions about how your code works
- **AI may not be used to generate code** — but you can use it to look up commands, examples, or circuits (verify anything you find against official sources)



For **Advanced Task** submission

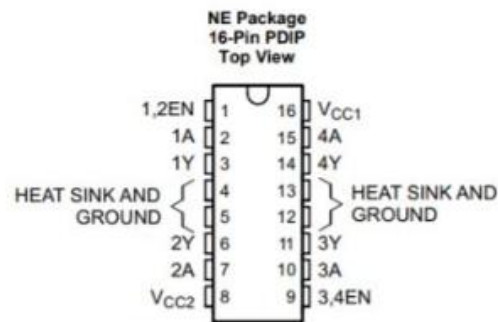
- Upload the **sketch file** and a **demo video** to GitHub — after your task check
- Create a "Lab 2" folder. Suggested naming: "Task2-1.ino", "Task2-1.mp4"
- Confirm with the lecturer or TA that your files uploaded successfully before you leave the classroom
- **AI tools** are allowed to help write your sketch — just be ready to explain how it works

Basic Task 2-1 Driving a DC Motor with L293D

🎯 Demo goal

Rotate the DC motor with Arduino.

Required components: DC motor, motor driver IC: L293D. Please refer to the figure below for L293D pin assignments.



Pin Functions

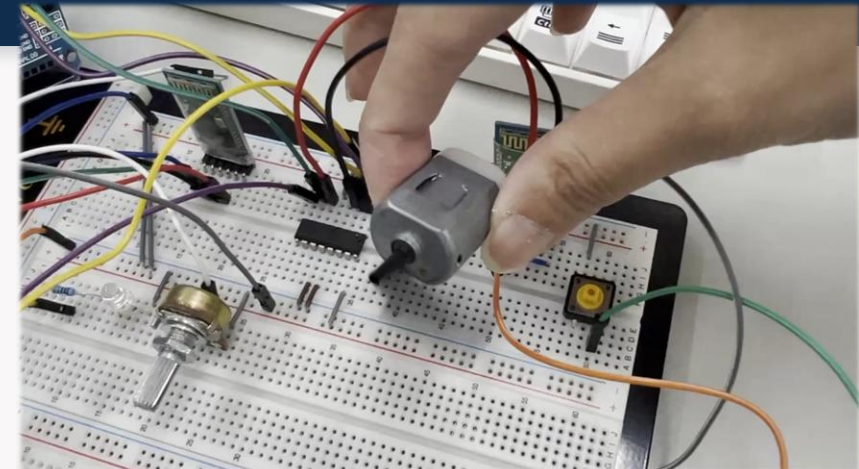
PIN		TYPE	DESCRIPTION
NAME	NO.		
1,2EN	1	I	Enable driver channels 1 and 2 (active high input)
<1:4>A	2, 7, 10, 15	I	Driver inputs, noninverting
<1:4>Y	3, 6, 11, 14	O	Driver outputs
3,4EN	9	I	Enable driver channels 3 and 4 (active high input)
GROUND	4, 5, 12, 13	—	Device ground and heat sink pin. Connect to printed-circuit-board ground plane with multiple solid vias
V _{CC1}	16	—	5-V supply for internal logic translation
V _{CC2}	8	—	Power VCC for drivers 4.5 V to 36 V

```
const int PIN_1A = ____;
```

```
const int PIN_2A = ____;
```

```
void setup() {  
    pinMode(____, ____);  
    pinMode(____, ____);  
}
```

```
void loop() {  
    digitalWrite(____, ____);  
    digitalWrite(____, ____);  
}
```



Basic Task 2-1 Notes



Explain & show during your demo

- How did you complete the circuit? If you used references, explain how you searched for them, verified whether the information was correct, and applied that knowledge to this task.
- Show the official L293D datasheet, open to the pin assignment page shown on the previous slide.



Safety

If the motor does not rotate, or there is a burning smell, please disconnect the Arduino from the power supply immediately!

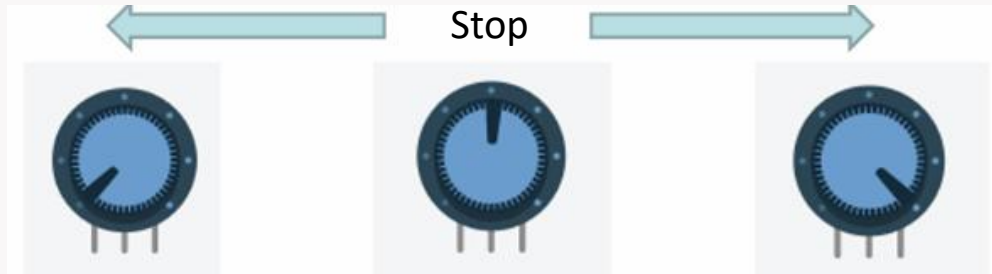
Advanced Task 2-1 Potentiometer Direction & Speed Control

🎯 Demo goal

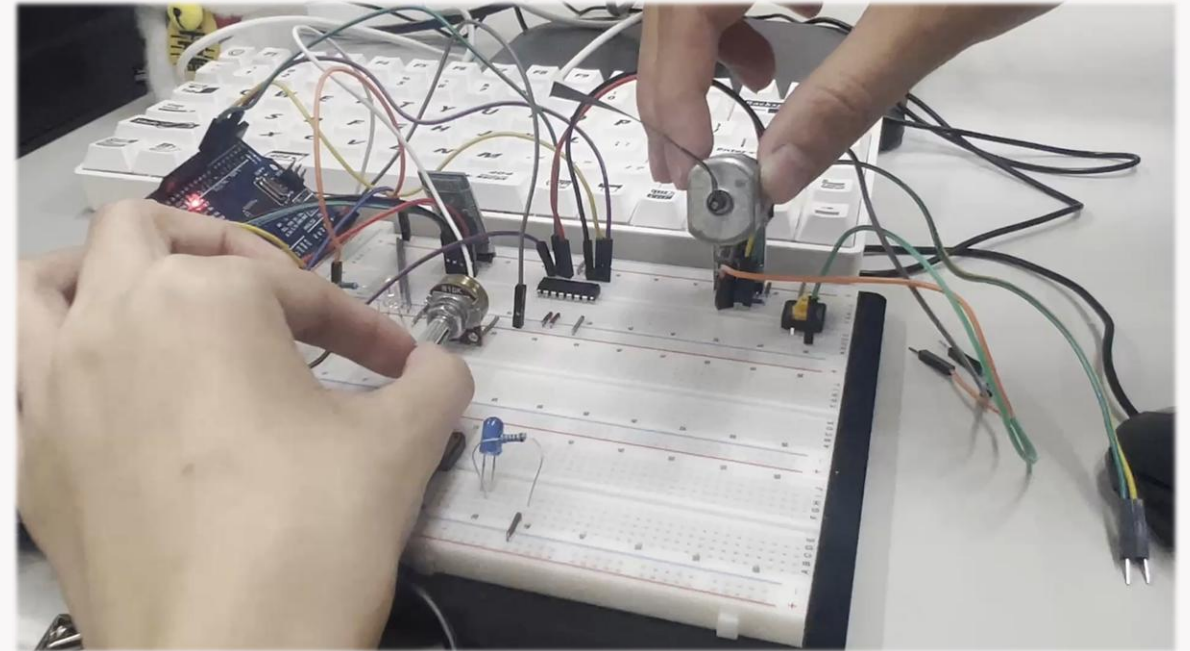
Add a variable resistor to the previous circuit, and use its angle to control both the motor's direction and speed.

Increased counterclockwise
rotation speed

Increased clockwise
rotation speed



Variable resistor



Basic Task 2-2 SG90 Servo Control

🎯 Demo goal

Controlling the angle of the servo motor SG90.

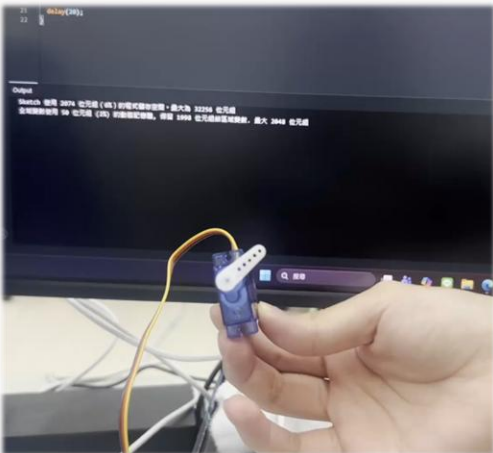
Complete the task by building the circuit and using the following sample code.

When the variable “data” is assigned a value “90”, the servo motor should be rotated to 90 degree once the sketch is uploaded to the Arduino.

Bonus Task: Complete this task by letting the user input the angle value from the Serial Monitor.

```
#include <Servo.h>
Servo myservo;
void setup()
{
  myservo.attach(____);
  myservo.write(____);
}

void loop() {
  int data = ____;
  myservo.write(____);
}
```



Hint

Please search by yourselves to learn about pin assignment of SG90 and how to use it.

Advanced Task 2-2 Ultrasonic Sensor Drives Servo Angle

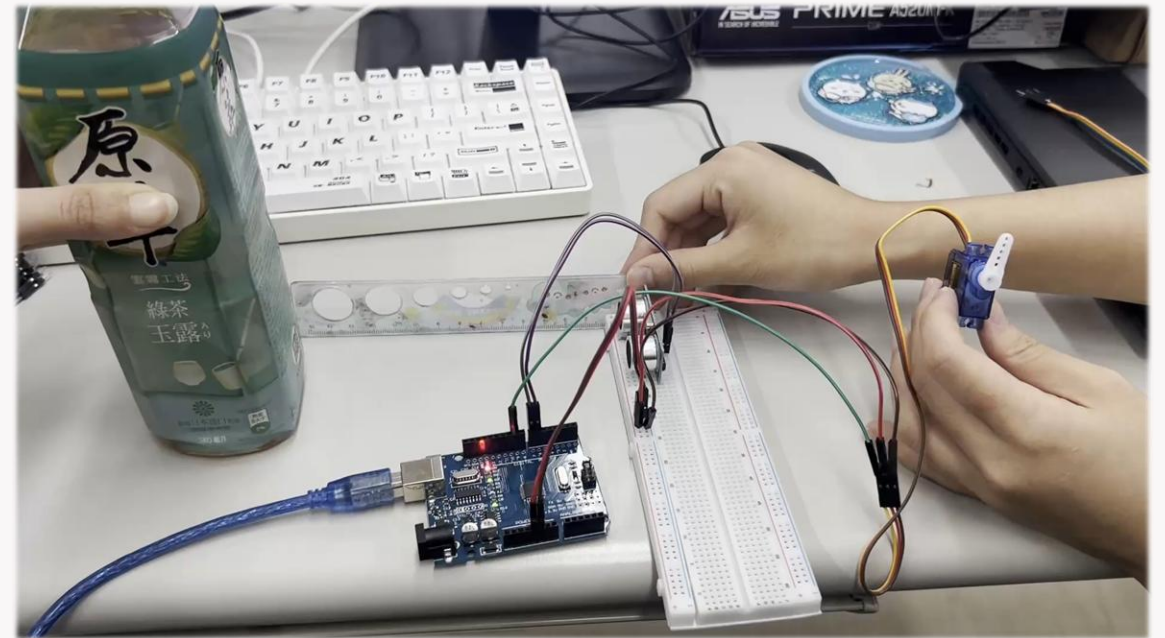
🎯 Demo goal

Servo Motor Control with an Ultrasonic Sensor
Controlling the angle of the servo motor according to the measured distance.

- Please investigate how to use the ultrasonic sensor module HC-SR04 by yourselves.

💡 Hint

Please search by yourselves to learn about pin assignment of SG90 and how to use it.



Basic Task 2-3 External Interrupt Controls an LED

Demo goal

Rebuild the push button (Button A) and LED (LED A) circuit in Basic Task 1-2. This time, use external interrupt to toggle the state of the LED by using the example sketch below.

Hint

- “`attachInterrupt`” and “`void buttonISR()`” functions are the keys to implement external interrupt. Investigate how to use them
- Pin assignment for external interrupt is critical! Only two IO pins for Arduino Uno support external interrupt

```
const int buttonPinA = __;
const int ledPinA = __;
volatile bool ledStateA = ____;

void setup ( ) {
    pinMode(ledPinA, ____);
    pinMode(buttonPinA, ____);
    attachInterrupt(____, ____, ____);
}

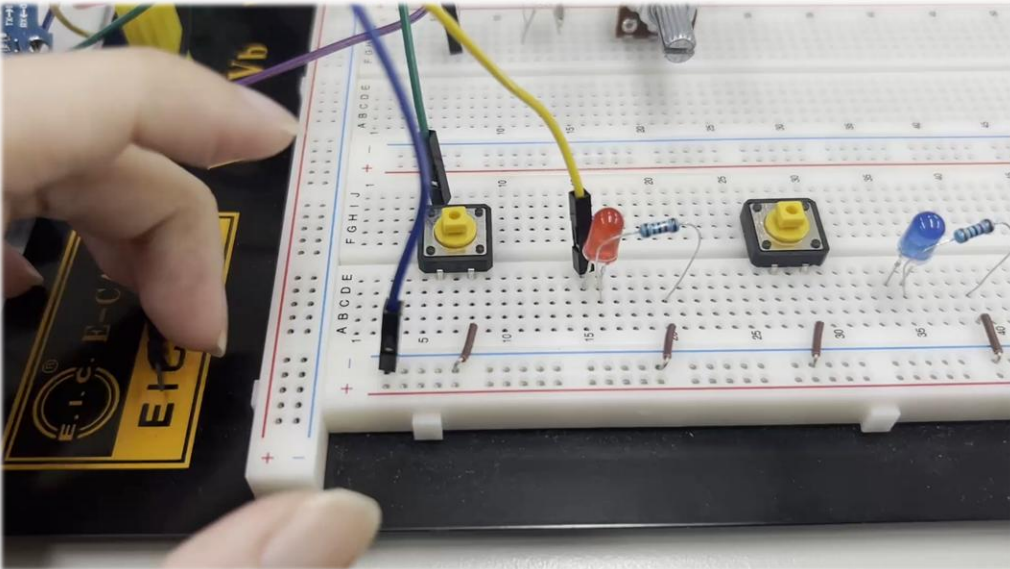
void buttonISR( ) {
    ledStateA = !ledState;
    digitalWrite(ledPinA, ledStateA);
}

void loop ( ) {
}
```

Basic Task 2-3 External Interrupt Controls an LED

? Why?

- What an external interrupt is
- Focus on the explanation of “`attachInterrupt`” and “`void buttonISR()`” functions, and their relation with the external interrupt



```
const int buttonPinA = __;
const int ledPinA = __;
volatile bool ledStateA = ____;

void setup ( ) {
    pinMode(ledPinA, ____);
    pinMode(buttonPinA, ____);
    attachInterrupt(____, ____, ____);
}

void buttonISR( ) {
    ledStateA = !ledStateA;
    digitalWrite(ledPinA, ledStateA);
}

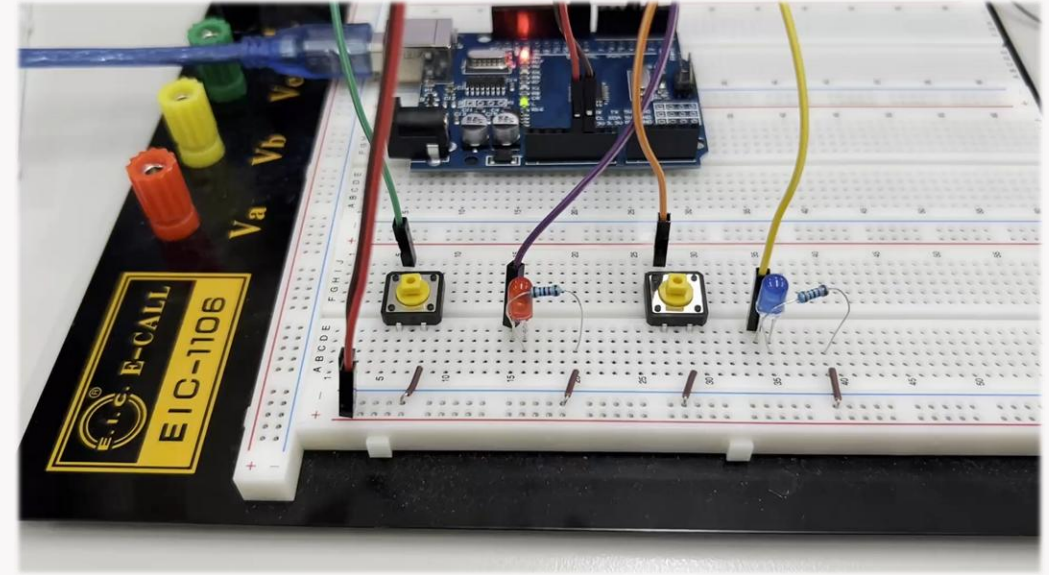
void loop ( ) {
}
```

Advanced Task 2-3 External Interrupt vs Polling

🎯 Demo goal

Build on Basic Task 2-3 by adding a second button + LED pair (Button B + LED B).

- **Button A + LED A:** keep using external interrupt to control the LED state
- **Button B + LED B:** use polling instead —
 - Continuously check Button B's state and detect press transitions (edge detection)
 - Toggle LED B when a press is detected
 - Add “`delay(2000)`” at the end of “`void loop()`” to simulate a busy/blocking system



❓ Why?

- What is polling? What are the differences between external interrupt and polling?
- Which method do you prefer to use for developing a robotic system?

Stuck? Troubleshooting Checklist



Motor won't spin at all

Check whether EN is connected and the truth-table logic is right



Servo angle gets stuck

Confirm `constrain()` is limiting the angle range



Ultrasonic readings look wrong

Distances outside the configured range produce unreasonable angles



Interrupt not responding

Confirm it's on pin 2 or 3, and `digitalPinToInterrupt()` is used

Lab 2 Complete!

Next → Lab 3: Timer Interrupts & Cross-Platform C# Communication

Q & A